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Effects of irradiance on the response of the coral *Acropora pulchra* and the calcifying alga *Hydrolithon reinboldii* to temperature elevation and ocean acidification

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ABSTRACT

We tested experimentally if irradiance can modulate the response of coral reef calcifiers to seawater warming and ocean acidification. Nubbins of the coral *Acropora pulchra* and individuals of the calcifying alga *Hydrolithon reinboldii* were incubated for 20 d under 2 irradiances (150 and 650 μmol quanta m^{-2} s^{-1}) in a matrix of duplicate treatments crossing 2 temperatures (27.2 °C and 29.8 °C) with 3 pCO_2 levels (400, 750 and 1100 μatm). To determine the effects of the treatments, net calcification was measured in *A. pulchra* and *H. reinboldii*, and biomass in *A. pulchra*. High temperature and low irradiance caused a significant decrease in coral net calcification, whereas only low irradiance resulted in a significant decrease in algal net calcification. The biomass of *A. pulchra* was affected significantly by pCO_2 and light (separately and in synergy), with maximum biomass measured at 750 μatm pCO_2 in 3 out of 4 combinations of light and temperature. Light intensity adds complexity to the response of reef calcifiers to ocean acidification through indirect effects on coral biomass, which will need to be considered in future studies.

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1. Introduction

Anthropogenic activities are affecting Earth by releasing large quantities of carbon dioxide (CO_2) into the atmosphere as a result of the burning of fossil fuels and deforestation (Sabine et al., 2004). As CO_2 is a greenhouse gas, it has driven a warming of the Earth's surface of 0.2 °C decade^{-1} for the past thirty years (Hansen et al., 2006). Oceans are affected by global warming since they have absorbed ~84% of the total heat of the Earth system over the last 40 years (Levitus et al., 2005). The International Panel on Climate Change (IPCC, 2007) projects for the end of the century an increase in ocean surface temperature of 1.8 to 4.0 °C depending on the scenario considered. Aside from global warming, ~25% of the human-produced CO_2 has been absorbed by seawater in the upper surface of the oceans (Sabine et al., 2004) leading to the "other CO_2 problem" known as ocean acidification (OA) (Gattuso and Hansson, 2011). OA is caused by the dissolution of CO_2 in seawater to form carbonic acid, leading to a decrease in ocean pH and a reduction in calcium carbonate saturation state (Ω).

In addition to a multitude of local disturbances (Pandolfi et al., 2003), coral reefs are threatened directly by global warming and OA (Gattuso and Hansson, 2011). Global warming leading to seawater warming is a major driver of mass bleaching episodes of tropical corals (Brown, 1997), which in a persistent state leads to reduced calcification, slow growth, and increased mortality (e.g., Glynn, 1996; Hoegh-Guldberg, 1999). Although, the effects of OA currently are less conspicuous than bleaching, the potential consequences are dire, with pessimistic projections suggesting that coral reefs will start dissolving by 2050 (Hoegh-Guldberg et al., 2007; Silverman et al., 2009). These projections are based in part on studies conducted in mesocosms showing that coral calcification declines ~20% when pCO_2 doubles (Chan and Connolly, 2013; Comeau et al., 2013a; Erez et al., 2011), and to date they have not evaluated whether at least some corals might show signs of resistance to OA as it has been shown in the field (Fabricius et al., 2011) and in short laboratory incubations (Comeau et al., 2013a). OA and global warming also are expected to threaten other important calcifiers on coral reefs, particularly coralline algae, which are significant contributors to the calcium carbonate budget of coral reefs (Chisholm, 2000). As it has been described for scleractinian corals, studies of coralline algae have detected deleterious effects of OA on growth, recruitment and calcification (e.g., Comeau et al., 2013a; Jokiel et al., 2008; Kuffner et al., 2008), although some species are affected less (e.g., Martin et al., 2013).

To date, most studies of the effects of OA and warming on reef calcifiers have tested the effects of these stressors independently, and of the few studies that have employed combinations of increased temperature and decreased pH, the results are equivocal and suggest that the responses are species-specific. For example, some studies report that temperature does not affect the response of corals to OA, including the temperate coral *Cladocora caespitosa* (Rodolfo-Metalpa et al., 2010) and the tropical coral massive *Porites* (Anthony et al., 2008; Edmunds et al., 2012). In other studies however, the interaction between

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increased temperature and high pCO_2 decreases calcification disproportionately as in the coral *Stylophora pistilla* (Reynaud et al., 2003), and reduces growth of primary polyps as in *Porites panamensis* (Anlauf et al., 2011). In tropical coralline algae, the interaction between warming and OA causes an increased sensitivity to bleaching, a decrease in net primary productivity, and a decrease in calcification (Anthony et al., 2008). In temperate coralline algae the effects of increasing pCO_2 and warming alone, and in combination, can be either positive or negative for calcification and photosynthesis as a function of the season (Martin and Gattuso, 2009; Martin et al., 2013).

Aside from seawater temperature and the concentration of dissolved inorganic carbon (DIC), coral and calcifying algal physiology also is strongly affected by irradiance, which drives photosynthesis and the supply of photosynthetically fixed carbon (e.g., Chalker et al., 1988; Muscatine, 1990). In symbiotic corals, the metabolic energy supplied to the host through the catabolism of photosynthetically fixed carbon, as well as the change in internal pH caused by photosynthesis (Allemand et al., 2011; Borowitzka and Larkum, 1987) favors calcification that is on average 3-fold higher in the light than in the dark (Gattuso et al., 1999). Similar to photosynthesis, calcification in corals is associated strongly with irradiance, and the relationship is best fitted with a hyperbolic tangent function (Chalker, 1981). Remarkably few studies have explored the combined effect of irradiance and pCO_2 on the calcification of corals, and the same is true for calcified algae, although it has long been known that their rates of calcification are light-enhanced (Borowitzka, 1981).

In corals, two studies have found no interactive effects of irradiance and pCO_2 (or Ω_a) on the calcification rates of adult *Porites compressa* and *Porites rus* (Comeau et al., 2013b; Marubini et al., 2001). In contrast, juvenile *Pocillopora damicornis* respond strongly to the interactive effect of irradiance and pCO_2, with the calcification–irradiance relationship following a hyperbolic tangent function under ambient pCO_2, but a reversed hyperbolic tangent function under high pCO_2 (Dufault et al., 2013). The inconsistency between the aforementioned studies likely is due to both differences between life stages and differences in the range of irradiances tested. For example, Marubini et al. (2001) used a range of irradiance from 81 to 698 μmol quanta m^{-2} s^{-1}, Comeau et al. (2013b) used 215 and 1000 μmol quanta m^{-2} s^{-1}, and Dufault et al. (2013) used 31 to 226 μmol quanta m^{-2} s^{-1}. A recent study also has shown that high pCO_2 induces a greater decrease in light calcification in corals incubated in low light compared to those incubated in high light, although dark calcification was similarly affected by high pCO_2 in the two light regimes (Suggett et al., 2013). This result supports the hypothesis that corals can use bicarbonate ions, which increase in concentration under OA conditions, to calcify in the light, thereby reducing the deleterious effects of decreasing carbonate ions which also is associated with OA (Comeau et al., 2013c).

In view of the critical role played by the three environmental parameters discussed above (temperature, pCO_2, and irradiance) on the rates of calcification in tropical corals and algae, our study tested the hypothesis that irradiance affects the response of corals and calcified algae to ocean warming and OA. To test this hypothesis, a 3-week incubation on the coral *Acropora pulchra* and the alga *Hydrolithon reinboldii* was performed, during which net calcification was measured under combinations of two light levels, two temperatures, and three pCO_2 levels. Biomass also was measured in the corals to investigate potential effects of treatments on the allocation of resources between tissue and skeleton growth.

2. Material and methods

2.1. Samples collection

Fragments of *A. pulchra* (n = 120) and individual *H. reinboldii* (n = 120) were collected in January 2013 in the lagoon of the North shore of Moorea, French Polynesia, at ~2 m depth. Samples were taken to the Richard B. Gump South Pacific Research Station, and attached to plastic tiles using epoxy glue (Z-Spar, A788 epoxy) to facilitate labeling and handling. After 2 d in a shallow tank supplied with a rapid flow of fresh seawater, samples were transferred to a recovery tank where they remained for 7 d. Lighting in the recovery tank was provided by four 75 W Light Emitting Diode (LED) modules (Sol White LED Module, Aquaillumination) that duplicated light quality and quantity used in the experiment (~150 and ~650 μmol quanta m^{-2} s^{-1} depending on the position in the tank). LEDs delivered most of the light in the spectrum 400–700 nm under the range of intensity utilized. Temperature in the recovery tank was maintained at ~28.5 °C, which was the seawater temperature at the collection site at the time of the experiment, and also was the mean of the temperatures used in the subsequent incubations (27.2 °C and 29.8 °C).

After recovery, samples were transferred to twelve 150 L mesocosms arranged randomly in a matrix of three pCO_2 and two temperature treatments that each were duplicated (see below). The experiment was a split-plot design, in which between-plot effects were pCO_2 and temperature, and the within-plot effect was irradiance that was manipulated in each mesocosm. The experiment was analyzed using analysis of variance (ANOVA). Five corals acclimated to low light and five acclimated to high light were selected randomly and transferred to the corresponding light treatment in each mesocosm. As it was not biologically meaningful to statistically compare taxa, corals and algae were incubated in the same mesocosms. Similarly, five algae acclimated to low light and five acclimated to high light were placed in the same mesocosms. The position of organisms in the mesocosms was changed randomly within a light level every 2 d to eliminate position effects, and seawater was replaced continuously in each mesocosm at ~200 mL min^{-1}.

The two light regimes (150 and 650 μmol quanta m^{-2} s^{-1} at maximum intensity) were created within each mesocosm by shading half of each mesocosm with neutral density mesh. The mesh attenuated ~77% of the irradiance and therefore created a shaded and unshaded portion in each mesocosm. To produce a natural diel light cycle, light intensity was gradually increased from 0 to 100% output over 4 h beginning at 06:00 h, maintained at maximum intensity for 4 h, and then reduced during the last 4 h of the 12-h day. To quantify the consistency of the light treatments, irradiance was measured once weekly below the seawater surface with a 4π quantum sensor (LI-193) and a LiCor LI-1400 meter in all 12 mesocosms during maximum light intensity. Irradiance also was measured in two different mesocosms during a 24 h period to quantify the natural light cycle created by altering the intensity of the lamps. The two irradiances were chosen with respect to photosynthetic performance to expose the organisms to saturating (650 μmol quanta m^{-2} s^{-1}) and light-limited (150 μmol quanta m^{-2} s^{-1}, e. g. Marubini et al., 2001) conditions. These irradiances are experienced on the shallow reefs of Moorea, where light intensity ranges from ~2000 μmol photons m^{-2} s^{-1} in the shallow (0.5 m depth) back reef to ~200 μmol photons m^{-2} s^{-1} in shaded areas of the back reef, and at ~17 m depth on the fore reef (Carpenter and Moorea Coral Reef LTER, 2012). Temperature was maintained at 27.2 °C in 6 of the 12 mesocosms, and at 29.8 °C in the remainder. These temperatures covered the range of temperatures recorded in the back reef of Moorea (e.g., Edmunds et al., 2010).

CO_2 treatments consisted of three pCO_2 levels corresponding to the present day concentration (400 μatm), an average value expected by the end of the century (~750 μatm, from representative concentration pathway [RCP] scenario 6.0, Moss et al., 2010), and a pessimistic value expected by the end of the 21st century (~1100 μatm, ~RCP scenario 8.5). The experiments were performed at constant pCO_2 because natural daily pCO_2 oscillations are limited on the back reef of the North shore of Moorea during this time of year (i.e., ~70 μatm, Comeau et al., in press-a). Either ambient air or CO_2-enriched air was bubbled into the tanks to create CO_2 treatments. A solenoid-controlled gas regulation system (Model A352, Qubit Systems) was used to create the CO_2-enriched air by mixing pure CO_2 and ambient air at the targeted value.

CO_2 levels within each mesocosm were monitored twice daily (see following section) and minor corrections of the gas flow and water flow were made for each mesocosm to ensure that actual pCO_2 was close to the targeted values.

2.2. Carbonate chemistry measurements

pH was measured twice daily using a portable pH meter (Orion 3-stars) with a DG 115-SC pH probe (Mettler) calibrated every other day with Tris buffer provided by Dr. Andrew Dickson (Scripps Institution of Oceanography). The pH probe was stable and drifted a mean of 0.1 mV between calibrations, which corresponded to <0.003 pH units. Twice during the incubations, pH_T also was measured spectrophotometrically in every mesocosm using the indicator dye m-cresol (Standard Operating Procedure 6b, Dickson et al., 2007). pH_T measured spectrophotometrically and with the pH probe differed ≤0.015 pH units. Total alkalinity (A_T) was measured every two days and remained stable within ≤10 μmol kg^{-1} during the 20-d incubation because of the open-flow of seawater. A_T titrations were performed on the day of seawater collection using open cell potentiometric titration with an automatic titrator (T50, Mettler-Toledo), and A_T calculated after Dickson et al. (2007). Titrations of certified reference material (batch 108) provided by Dr. A. Dickson were performed prior to each set of titrations and yielded an average value of 2221 ± 3.1 μmol kg^{-1} (±SE, n = 10) for a certified value of 2218 μmol kg^{-1}. Parameters of the carbonate system were calculated from salinity, temperature, A_T, and pH_T using the R package seacarb (Lavigne and Gattuso, 2012).

2.3. Net calcification

Buoyant weights of the corals and algae (Spencer-Davies, 1989) were recorded at the beginning of the incubation and after 20 d in the treatments with a precision of ±1 mg. The difference between the initial and final buoyant weights was converted to dry weight using an aragonite density of 2.93 g cm^{-3} for *A. pulchra*, and a calcite density of 2.71 g cm^{-3} for *H. reinboldii*. Rates of net calcification were normalized to the area of organisms estimated using aluminum foil (Marsh, 1970) for *A. pulchra*, and by wax dipping (Stimson and Kinzie, 1991) for *H. reinboldii*. Biomass also was measured for *A. pulchra* to investigate the potential impact of light, temperature, and pCO_2 on tissue resources. Corals were fixed in 10% formalin for 48 h, then placed in 5% HCl until the skeleton dissolved (1–2 d). The tissue tunics were rinsed in distilled water and dried for 48 h at 60 °C prior to weighing (±1 mg), and normalized to area (mg cm^{-2}).

2.4. Statistical analyses

Temperature and pCO_2 conditions in the mesocosms were analyzed with a two factor nested ANOVA in which treatment (pCO_2 or temperature) was a fixed effect and mesocosms a random nested factor. Area-normalized net calcification and biomass were analyzed with a four-way mixed model ANOVA in which temperature and pCO_2 were fixed between plot effects, duplicate mesocosms were a random effect nested in each treatment, and irradiance was a fixed, split-plot effect in each mesocosm. When the mesocosm effect was not significant ($p \geq 0.25$) (Quinn & Keough, 2002) it was dropped from the statistical analyses, and the net calcification and biomass were re-analyzed using a three-way ANOVA in which pCO_2 and temperature were fixed, between-plot effects, and irradiance was a fixed split-plot effect. The assumptions of normality and equality of variance were evaluated through graphical analyses of residuals. All analyses were performed using the software R (R Foundation for Statistical Computing).

3. Results

Temperature and carbonate chemistry conditions were maintained precisely in the treatments (Table 1), with mean pCO_2 treatments of 402 ± 4, 718 ± 12 and 1107 ± 22 μatm (mean ± SE; n = 72) and mean temperature treatments of 27.2 ± 0.07 and 29.8 ± 0.08 °C (mean ± SE; n = 72). pCO_2 treatments were not significantly different among duplicate mesocosms ($p = 0.877$), but differed significantly between treatments ($p < 0.001$). Temperature also did not differ among mesocosms in similar treatments ($p = 0.640$), but differed between treatments ($p < 0.001$). The light treatment contrasted 639.9 ± 3.0 μmol quanta m^{-2} s^{-1} (mean ± SE; n = 240) in the high light treatment versus 149.2 ± 0.1 μmol quanta m^{-2} s^{-1} (mean ± SE; n = 240) in the low light treatment, as measured during the 4 h of maximum intensity (from 10:00 h to 14:00 h) (Fig. 1). Due to the ramping periods at the start and end of each day, the average light levels experienced during the 12 h day were ~411.4 ± 8.6 μmol quanta m^{-2} s^{-1} in the high light treatment, and 95.8 ± 2.1 μmol quanta m^{-2} s^{-1} in the low light treatment (both ± SE n = 720).

For corals, no mortality or bleaching occurred during the 20 d treatments, and area-normalized net calcification was positive in all the treatments (Fig. 2A). The highest net calcification rate was measured at 27.2 °C in the high light, 400-μatm treatment (1.02 ± 0.05 mg $CaCO_3$ d^{-1} cm^{-2}, [mean ± SE; n = 10]), and the lowest net calcification rates were found in the 27.2 °C, low-light treatment at both 400-μatm and 1100-μatm (0.26 ± 0.03 mg $CaCO_3$ d^{-1} cm^{-2} and 0.25 ± 0.04 mg $CaCO_3$ d^{-1} cm^{-2} respectively). The ANOVA showed that net calcification was not affected significantly by pCO_2 ($p = 0.093$, Table 2). An exploratory post hoc analysis of the corals maintained under high light alone demonstrated that net calcification decreased significantly as a function of increasing pCO_2 under this irradiance ($F_{2,54} = 6.23$, $p = 0.04$). There was no significant interaction between pCO_2 and temperature ($p = 0.546$), but temperature alone significantly affected net calcification ($p = 0.050$, Table 2). Light significantly affected net calcification ($p < 0.001$, Table 2) with lower rates under reduced light (Fig. 2A), but there were no interactive effects of light and temperature ($p = 0.831$, Table 2), or light and pCO_2 ($p = 0.086$, Table 2).

Area-normalized biomass of *A. pulchra* was highest at 27.2 °C in the high light 750-μatm, where it was 3.15 ± 0.29 mg cm^{-2}, and lowest at 29.8 °C and high light 400-μatm, where it was 1.87 ± 0.18 mg cm^{-2} (±SE, n = 10). Biomass was significantly affected by pCO_2 ($p = 0.028$) and temperature ($p = 0.041$), but not by the interaction between the two ($p = 0.517$) (Table 2). Light and the interaction between light and pCO_2 significantly affected the biomass ($p = 0.033$ and 0.009 respectively, Table 2). Under high light at both temperatures, biomass was maximum at intermediate pCO_2 (750 μatm), where it was 31% and 38% higher than at ambient pCO_2 at 29.8 °C and 27.2 °C, respectively. Biomass also was maximum at intermediate pCO_2 (750 μatm) under low light at 27.2 °C, where it was ~10% higher than at ambient pCO_2. In contrast, under low light and at 29.8 °C, biomass was similar in all pCO_2 treatments.

For the alga *H. reinboldii*, no mortality occurred during the 20-d incubation, and net calcification was positive in all treatments (Fig. 3). The highest net calcification was measured in the high light, 400-μatm pCO_2 treatments at 27.2 °C (0.125 ± 0.018 mg $CaCO_3$ d^{-1} cm^{-2}, [mean ± SE; n = 10]) and 29.8 °C (0.126 ± 0.015 mg $CaCO_3$ d^{-1} cm^{-2}). The lowest net calcification was measured in the 29.8 °C low-light treatment at 1100-μatm pCO_2 (0.045 ± 0.019 mg $CaCO_3$ d^{-1} cm^{-2}). pCO_2 did not significantly affect area-normalized net calcification ($p = 0.086$, Table 3, Fig. 3). When pCO_2 increased from 400-μatm to 1100-μatm, the net calcification decreased between 17% and 52% at high-light and high temperature (29.8 °C) and at low-light and ambient temperature (27.2 °C), respectively. Temperature, as well as the pCO_2 × temperature interaction, did not affect net calcification ($p = 0.968$ and $p = 0.725$ respectively, Table 3). Light significantly affected net calcification ($p = 0.048$,

Table 1
Mean carbonate chemistry in the 12 incubation tanks during the 20-d of incubation. Mean ± SE pH_T (n = 72), total alkalinity (A_T) (n = 18), temperature (n = 72), and salinity (n = 5) were used to calculate pCO_2, the dissolved inorganic carbon concentration (DIC), and the saturation state of aragonite (Ω_{arag}) and calcite (Ω_{calc}) using the R package Seacarb (Lavigne and Gattuso, 2012). SE are <0.01 for pH_T.

Condition	pH_T	A_T (μmol kg^{-1})	DIC (μmol kg^{-1})	pCO_2 (μatm)	Ω_{arag}	Ω_{calc}	Salinity	Temperature (°C)
27 °C–400 μatm	8.06	2313 ± 2	1975 ± 2	379 ± 3	3.78 ± 0.02	5.69 ± 0.03	35.3 ± 0.1	27.3 ± 0.04
27 °C–750 μatm	7.84	2311 ± 1	2101 ± 3	711 ± 9	2.49 ± 0.02	3.76 ± 0.04	35.2 ± 0.1	27.1 ± 0.02
27 °C–1100 μatm	7.67	2316 ± 1	2182 ± 3	1108 ± 15	1.79 ± 0.02	2.70 ± 0.03	35.3 ± 0.1	27.1 ± 0.05
30 °C–400 μatm	8.03	2317 ± 2	1971 ± 3	405 ± 3	3.92 ± 0.02	5.86 ± 0.03	35.3 ± 0.1	29.8 ± 0.05
30 °C–750 μatm	7.83	2315 ± 1	2089 ± 3	726 ± 8	2.71 ± 0.03	4.04 ± 0.04	35.3 ± 0.1	29.9 ± 0.03
30 °C–1100 μatm	7.67	2318 ± 2	2167 ± 3	1107 ± 16	1.99 ± 0.02	2.97 ± 0.03	35.4 ± 0.1	29.8 ± 0.05

Table 3), with a 29% lower mean net calcification in low light, but there were no significant interactions of light with temperature or pCO_2.

4. Discussion

4.1. Effects on the coral A. pulchra

By measuring both net calcification and biomass of corals, we were able to show that these two parameters are not affected in a similar manner by changing environmental factors. The effect of pCO_2 on net calcification was not statistically significant, although net calcification was reduced at higher pCO_2 under high light intensity, particularly at 27.2 °C. Similar to the present study, previously we have reported a linear decline in net calcification in *A. pulchra* maintained at 27.2 °C and high light intensity (700 μmol photons m^{-2} s^{-1}) under an array of pCO_2 values ranging from 280 μatm to 2100 μatm (Comeau et al., 2013a). In the present study, net calcification decreased ~14% when pCO_2 was doubled under high light and 27.2 °C, whereas in Comeau et al. (2013a) we found under similar conditions a decrease of ~9%. Such differences could reflect a seasonal difference in the magnitude of the effect of pCO_2 on net calcification (Fitt et al., 2000; Thornhill et al., 2011), since our previous study (Comeau et al., 2013a) was carried out in September (when the temperature of ambient seawater was lower [~27 °C] and coral biomass was likely near maximum [Fitt et al., 2000; Thornhill et al., 2011]), whereas the present study took place in February (when ambient seawater was warmer [28.5 °C] and coral biomass was likely near minimum [(Fitt et al., 2000; Thornhill et al., 2011)]). Although previously we maintained light intensity at a constant of 700 μmol quanta m^{-2} s^{-1} in a 12:12 h light:dark cycle (Comeau et al., 2013a), ramping periods for light intensity were used in the present study to reproduce natural lighting conditions, which also could explain why the present results differ slightly from our previous work with *A. pulchra*.

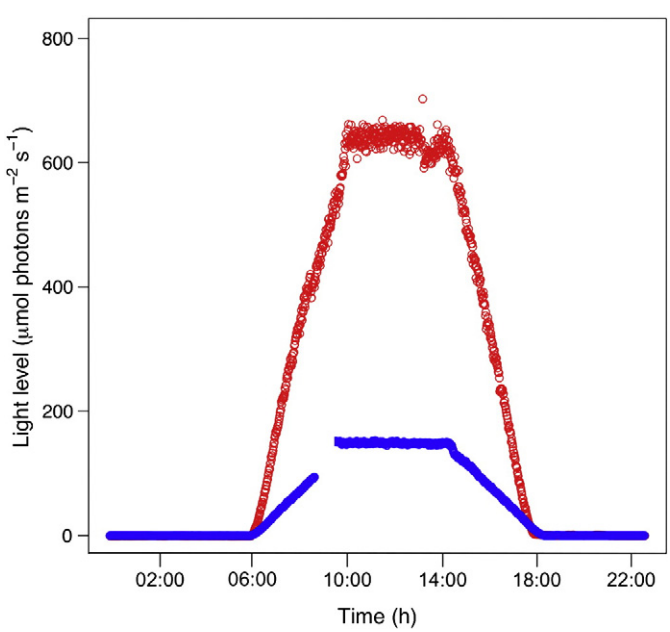

Fig. 1. Light levels measured during a 24 h cycle in one of the mesocosms. The two light regimes targeting 150 and 650 μmol quanta m^{-2} s^{-1} were obtained by shading half of each mesocosm using shade cloth attached to the mesocosm lids.

In contrast to pCO_2, temperature significantly affected coral net calcification with slightly lower net calcification at 29.8 °C compared to 27.2 °C. The strongest effect of temperature was seen at 400-μatm pCO_2, where high temperature (29.8 °C) reduced net calcification by 18% and 50% under high and low light, respectively. In contrast, at 750 μatm pCO_2 the differences in area-normalized net calcification between the 2 temperatures were ~10%. Similar higher sensitivity to temperature at ambient pCO_2 has been found for *Acropora intermedia*, for which increasing temperature (28–29 °C) reduced net calcification at ambient seawater pH (8.00–8.40) but had no effect on net calcification at pH of 7.85–7.95 (equivalent to our 750-μatm pCO_2 treatment) (Anthony et al., 2008). These results contrast with findings for *P. rus* where increasing temperature (25.6 °C up to 29.3 °C) led to a ~40% increase in net calcification at ambient pCO_2 (~420 μatm), and a ~20% increase in net calcification at high pCO_2 (~820 μatm) (Edmunds et al., 2012). Edmunds et al. (2012) and Anthony et al. (2008) also found that net calcification rates of massive *Porites* spp. were unaffected by the interaction between temperature and pCO_2. The aforementioned results suggest that the response of corals to OA and warming is species-specific and driven mostly by temperature. In species displaying reduced sensitivity to OA and warming such as massive *Porites* (Fabricius et al., 2011; Loya et al., 2011), net calcification appears to be unaffected by a combination of the 2 stressors. In *P. rus* and *Acropora*, net calcification is driven mostly by temperature, with OA mitigating the temperature effect. At control (i.e., ambient) pCO_2, increased temperature affected net calcification in a positive (*P. rus*) or negative (*Acropora*) way, but when the temperature increase was accompanied by an increase in pCO_2, the effect of warming was reduced.

As expected from the well-known light-dependence of calcification in tropical corals (e.g., Chalker and Taylor, 1978), in the present study irradiance was the main driver of net calcification, with mean net calcification of corals (pooled among pCO_2 and temperature treatments) reduced by 59% at low light compared to high light. This result is consistent with measurements made by Chalker (1981) that showed a decrease of ~70% in net calcification when light was decreased from 650 to 150 μmol photons m^{-2} s^{-1} for three species of coral. In our study irradiance did not interact with temperature and pCO_2 to affect net calcification, which is in agreement with two studies that have found no interactive effects of light and pCO_2 (or Ω_a) on the net calcification of *P. rus* and *P. compressa* (Comeau et al., 2013b; Marubini et al., 2001). In contrast, strong interactive effects of light and pCO_2 have been reported for recruits of *P. damicornis* with strong differences in net calcification between samples exposed to light level of ~70 μmol quanta m^{-2} s^{-1} at 493 versus 878 μatm pCO_2 (Dufault et al., 2013). In addition to the difference in life stages (recruits vs. adults), the difference in light intensities used by Dufault et al. (2013) (from 31 to 226 μmol quanta m^{-2} s^{-1}) and the

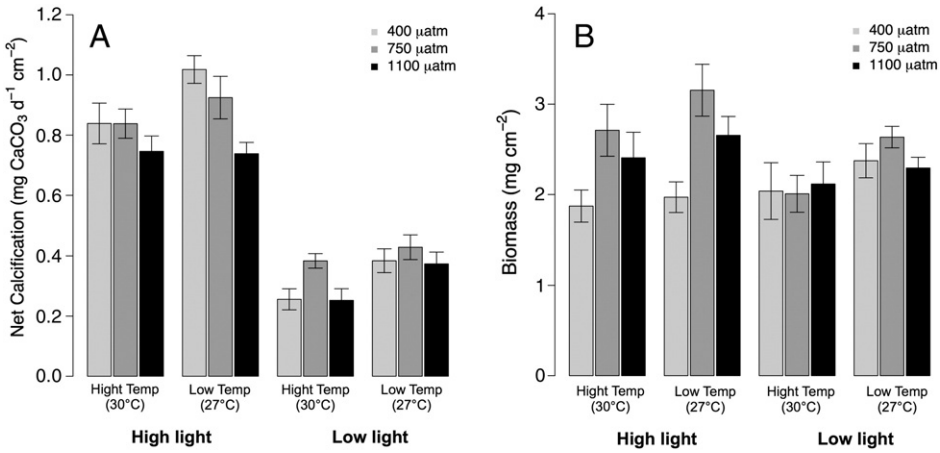

Fig. 2. Combined effects of two light levels (650 μmol quanta m^{-2} s^{-1} and 150 μmol quanta m^{-2} s^{-1}), two temperatures (27.2 °C and 29.8 °C) and 3 pCO_2 (400, 750 and 1100 μatm) on the coral *Acropora pulchra*. The effects were investigated on: (A) the surface area-normalized net calcification rates over 20 d, and (B) the biomass per surface unit measured at the end of the incubation. Values displayed are mean ± SE (n = 10).

present study (150 and 650 μmol quanta m^{-2} s^{-1}) is likely to explain the dissimilarities between the studies. In Dufault et al. (2013), the highest combined effect of light and pCO_2 on recruits of *P. damicornis* was measured at a light intensity below the one used in the present study (between 50 and 100 μmol quanta m^{-2} s^{-1}). The interactive effect of light and pCO_2 detected in Dufault et al. (2013) (but not in the present study) is likely to be ecologically significant, because coral larvae often settle in cryptic locations (Birkeland et al., 1981), where low light intensities would accentuate pCO_2 effects on recruits (Dufault et al., 2013).

The ways in which tissue biomass in reef corals changes in quality and quantity in response to global climate change and OA has received little attention compared to calcification. This may be an important oversight given evidence that tissue thickness in corals is related to the size of energy reserves and the capacity for photo-protection of *Symbiodinium* (Edmunds, 2011; Fitt et al., 2000; Loya et al., 2011; Thornhill et al., 2011), and may be related directly to the capacity to resist the effects of OA (Edmunds, 2011). The combined effects of pCO_2 and temperature on biomass have been studied in massive *Porites* spp. exposed for 1 month to 416 and 816 μatm pCO_2 at 25.7 °C and 29.3 °C (Edmunds, 2011). In massive *Porites* spp. deprived of zooplankton (comparable to the corals in the present study), biomass was unaffected by pCO_2 at low temperature, but decreased when both pCO_2 and temperature were elevated. Recently, we have studied the effects of light intensity (211 and 1015 μmol quanta m^{-2} s^{-1}) and pCO_2 (387 versus 719 μatm) on the tissue biomass of *P. rus*, maintained in conditions similar to those used in the present study, and did not find interactive effects of light and pCO_2 (Comeau et al., 2013b).

In the present study, biomass of *A. pulchra* was depressed 13% at high temperature and 9% at low light, whereas pCO_2 affected biomass in a non-linear fashion with a maximum stimulatory effect at 750 μatm in 3 out of 4 combinations of irradiance and temperatures. In the low light and high temperature treatment, biomass was constant across pCO_2 levels. We have described a similar non-linearity of the response of biomass to pCO_2 for both *A. pulchra* and *P. rus* in which biomass exhibited a positive parabolic relationship with pCO_2 with a threshold at ~1000 μatm (Comeau et al., 2013a). These results are consistent with the parabolic relationship between photosynthetic production and pCO_2 described in *Acropora formosa* (Crawley et al., 2010), and suggest that DIC is limiting for photosynthesis of *Symbiodinium* within coral tissues at ambient pCO_2 (Marubini et al., 2008). With DIC limitation, *Symbiodinium* photosynthesis then benefits from the increased availability of HCO_3^- and CO_2 associated with OA, at least to a threshold level of pCO_2. When pCO_2 exceeds ~750–1000 μatm, physiological processes such as enzymatic kinetics may be functionally impaired by the larger declines in pH associated with high pCO_2 (Pörtner et al., 2004), as is the case for the enzyme phosphoglycolate phosphatase in the *Symbiodinium* of *A. formosa* (Crawley et al., 2010).

Experimental analyses of photophysiological performance were beyond the scope of the present study, but such studies may be important to provide pieces of the puzzle leading to a detailed understanding of

Table 2
Comparison of area-normalized net calcification and biomass among treatments for *Acropora pulchra*. Area-normalized net calcification and biomass were analyzed with a four-way, mixed model ANOVA in which temperature and pCO_2 were fixed between plot effects, duplicate mesocosms served as a random effect nested in each treatment, and light was a fixed, split-plot effect in each mesocosm. Since the mesocosm effect was not significant ($p \geq 0.25$), it was dropped from the statistical analysis, and net calcification and biomass were re-analyzed using a three-way ANOVA in which pCO_2 and temperature were fixed, between-plot effects, and light was a fixed split-plot effect in each mesocosm.

Dependent variable	Source	Effect	df	MS	F	p-Value
Area-normalized net calcification (mg $CaCO_3$ d^{-1} cm^{-2})	Between-plots	pCO_2	2	0.155	3.611	0.093
		Temperature	1	0.255	5.944	0.050
		pCO_2 × temperature	2	0.029	0.671	0.546
	Within-plots	Light	1	7.63	333.16	<0.001
		Light × pCO_2	2	0.087	3.787	0.086
		Light × temperature	1	0.001	0.050	0.831
		Light × pCO_2 × temperature	2	0.026	1.119	0.386
		Residuals	96	0.023		
Biomass (mg cm^{-2})	Between-plots	pCO_2	2	3.180	6.852	0.028
		Temperature	1	3.110	6.700	0.041
		pCO_2 × temperature	2	0.342	0.738	0.517
	Within-plots	Light	1	1.415	7.646	0.033
		Light × pCO_2	2	2.094	11.314	0.009
		Light × temperature	1	0.102	0.554	0.485
		Light × pCO_2 × temperature	2	0.068	0.366	0.708
		Residuals	96	0.533		

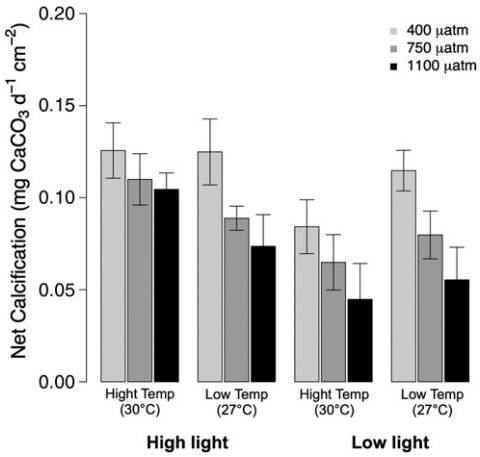

Fig. 3. Combined effects of two light levels (650 μmol quanta m^{-2} s^{-1} and 150 μmol quanta m^{-2} s^{-1}), two temperatures (27.2 °C and 29.8 °C) and 3 pCO_2 (400, 750 and 1100 μatm) on the surface area-normalized net calcification rates of the alga *Hydrolithon reinboldii*. Net calcification rates were measured over 20 d. Values displayed are mean ± SE (n = 10).

the mechanisms underlying the response of biomass and calcification in corals to combination of pCO_2 and irradiance. Only a better understanding of the mechanism of coral calcification, and notably how it is tightly coupled with photosynthesis, will provide accurate predictions of the response of corals to OA. Future efforts will need to focus on establishing the relationship between calcification and irradiance for corals incubated under a large range of ecologically relevant irradiances and crossed with a range of pCO_2 levels representative of past, present, and future reef conditions (Dupont and Pörtner, 2013).

Our study shows a difference in the response of calcification and biomass in a coral to the treatments tested. The most conspicuous differences in these response variables occurred under low light intensity, where net calcification was enhanced by increasing pCO_2 to a threshold level, whereas biomass remained constant (at 29.8 °C) or slightly increased (at 27.2 °C). These results indicate that the potentially beneficial effects of increased rates of photosynthesis at high pCO_2 (i.e., larger quantities of photosynthetically fixed carbon), even at sub-saturating light intensities (Crawley et al., 2010), may be used to increase net calcification and not biomass under low light. In contrast, at high light intensity, the beneficial effect of increasing pCO_2 (up to 750 μatm) for photosynthesis is reflected in increased biomass (~35%), whereas net calcification is unaffected (29.8 °C) or reduced (27.2 °C).

4.2. Effects on the alga H. reinboldii

For *H. reinboldii*, net calcification was not significantly affected by pCO_2, although net calcification decreased as pCO_2 increased in all combinations of temperature and light. In a recent study we have shown that when exposed to a large range of pCO_2 (6 levels from 280 μatm up to 2100 μatm), net calcification of *H. reinboldii* decreased linearly as a function of increasing pCO_2 (Comeau et al., in press-b). In contrast to corals, temperature alone or in combination with pCO_2 and light did not affect net calcification in *H. reinboldii*. Results from other studies indicate a species-specific response of coralline algae to the combination of increasing temperature and pCO_2. In the crustose coralline alga *Porolithon onkodes* temperature and pCO_2 have been described to act in synergy to affect growth (calcification), net productivity, and bleaching (Anthony et al., 2008). Whereas Anthony et al. (2008) described bleaching (i.e., adoption of a pale color) in *P. onkodes* at high pCO_2 (1010–1360 μatm), we did not observe bleaching. The effects of temperature (ambient and ambient + 3 °C) and pCO_2 (400 and 700 μatm) also have been studied on the temperate coralline alga *Lithophyllum cabiochae* (Martin and Gattuso, 2009), which was unaffected by each factor in isolation, but showed a 50% reduction in net calcification (over a whole summer) when exposed to a combination of high temperature and high pCO_2. The reduced effect of temperature described in our study could be due to acclimatization of *H. reinboldii* to the range of temperatures used in the experiment, because on the back reef of Moorea, seawater temperature routinely approaches (and sometimes exceeds) 29.8 °C during the austral summer (Edmunds et al., 2010).

As described above for *A. pulchra*, in *H. reinboldii* light intensity was the main factor driving calcification, with rates 29% greater in high compared to low light irradiance. This result is in agreement with previous measurements of net calcification made for the same alga under different irradiances (Chisholm, 2000), which revealed, as for corals, a hyperbolic relationship between light and net calcification with maximum net calcification at light intensities >500 μmol quanta m^{-2} s^{-1}. It also has been shown that light affects the net calcification rate of the Arctic coralline algae *Lithothamnion glaciale* maintained under different pCO_2 levels. In the light (i.e., the Arctic summer) *L. glaciale* was less susceptible to dissolution at low saturation states than in darkness (i.e., the Arctic winter) (Büdenbender et al., 2011), which likely is due to the enhancement of $CaCO_3$ precipitation in the light with constant rates of $CaCO_3$ dissolution.

5. Conclusions

Our results provide indications of the ways in which *A. pulchra* and *H. reinboldii* will respond to rising pCO_2 and elevated temperature in reef environments differing in light intensity (as might occur between seasons, depths, or microhabitats). First, in both taxa, there was no positive synergy between high temperature and high pCO_2 in depressing rates of net calcification, and therefore these taxa are unlikely to be disproportionately affected by future seas that are simultaneously warmer and more acidic than contemporary seas. Second, biomass or net calcification increased in a light-dependent manner in *A. pulchra* exposed to 750 μatm pCO_2. One likely cause for this effect is through enhanced photosynthesis of *Symbiodinium* caused by increased HCO_3^- at high pCO_2, with this effect cascading over to affect host through translocated photosynthetically fixed carbon. Third, in the two species we studied,

Table 3
Comparison of area-normalized net calcification among treatments for *Hydrolithon reinboldii*. Area-normalized net calcification was analyzed with a four-way mixed model ANOVA in which temperature and pCO_2 were fixed between plot effects, duplicate mesocosms were treated as a random effect nested in each treatment, and light was a fixed, split-plot effect in each mesocosm. Since the tank effect was not significant ($p \geq 0.25$), it was dropped from the statistical analysis, and net calcification was re-analyzed using a three-way ANOVA in which pCO_2 and temperature were fixed, between-plot effects, and light was a fixed split-plot effect in each mesocosm.

Dependent variable	Source	Effect	df	MS	F	p-Value
Area-normalized net calcification ($mg\ CaCO_3\ d^{-1}\ cm^{-2}$)	Between-plots	pCO_2	2	0.019	3.805	0.086
		Temperature	1	$9.0 \cdot 10^{-6}$	0.002	0.968
		pCO_2 × temperature	2	$1.7 \cdot 10^{-3}$	0.339	0.725
	Within-plots	Light	1	0.028	6.152	0.048
		Light × pCO_2	2	$5.3 \cdot 10^{-4}$	0.116	0.892
		Light × temperature	1	$9.8 \cdot 10^{-3}$	2.154	0.193
		Light × pCO_2 × temperature	2	$6.6 \cdot 10^{-5}$	0.014	0.986
		Residuals	96	0.005		

low irradiance (150 μmol quanta m^{-2} s^{-1}) did not accentuate the negative effects of OA on calcification we detected at high light intensity, as might be expected from recent work on coral recruits (Dufault at al., 2013). Therefore in more acidic future seas and relative to present day rates of area-normalized net calcification, *A. pulchra* and *H. reinboldii* in shaded microhabitats will not be more disadvantaged compared to their conspecifics in high light environments. Finally, regardless of the nuanced effects described above, the decrease in net calcification measured as a function of increasing pCO_2 in *H. reinboldii* and *A. pulchra* under high light intensity (and low light intensity for *H. reinboldii*), and at both investigated temperatures, confirms that some species of coral reef calcifiers will be impacted by decreasing pH independent of irradiance or rising seawater temperatures associated with global warming.

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